

DIP JOTI PAUL

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EDUCATION

Massachusetts Institute of Technology (MIT), Massachusetts, USA

Ph.D. Candidate, Electrical Engineering and Computer Science 01/2021 – 10/2026 (expected)

Advisor: Prof. Karl Berggren

Research topics: Photonics, Superconducting detectors and electronics, Nanofabrication

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

M.Sc., Electrical and Electronic Engineering (EEE) CGPA: 4.0/4.0 08/2020

B.Sc., Electrical and Electronic Engineering (EEE) Class rank: 3/194 03/2017

TECHNICAL SKILLS

- Cryogenic Measurements: Superconducting device characterization (300 mK–4 K): single-photon detectors, superconducting nanowire electronics, kinetic inductance, cryogenic probe station.
- Programming: Python (PyVISA, SCPI-based instrument automation), C++.
- Circuit Design and Simulation: LTspice, Cadence Virtuoso, Silvaco TCAD, Verilog.
- Multiphysics and EM Simulation: COMSOL Multiphysics, Ansys Lumerical FDTD.
- Nanofabrication: Electron-beam lithography, photolithography, dry etching, sputtering, ALD, e-beam evaporation, AFM, SEM, FTIR, ellipsometry, and XRD.

HONORS and ACHIEVEMENTS

- [IEEE Council on Superconductivity \(CSC\) Graduate Fellowship \(2026\)](#)
- [2025-2026 MathWorks Fellow](#), MIT School of Engineering
- MIT RLE Claude E. Shannon Fellowship (2025); featured in [Weston Observer](#)
- Honorable Mention, APS Division of Laser Science Student Poster Competition, CLEO 2022
- Dean's List for Academic Excellence, each undergraduate year (BUET)

SELECTED PUBLICATIONS

1. **D.J. Paul**, S. Koppell, G.G. Taylor, B. Korzh, S.R. Patel, A.D. Beyer, E.E. Wollman, M.D. Shaw, P. D. Keathley, and K.K. Berggren, "Enhanced Mid-Infrared Single-Photon Detection with Antenna-Coupled Superconducting Nanowires", [Nano Letters](#) (2026).
2. F. Incalza, M. Castellani, **D.J. Paul**, A. Simon, E. Batson, D. Mondin, O. Medeiros, and K.K. Berggren, "Fast-Recovery Epitaxial NbN Superconducting Nanowire Single-Photon Detectors with Saturated Efficiency at 1550 nm in Liquid Helium", [Nano Letters](#) (2026).
3. S. Koppell*, O. Bittencourt*, **D.J. Paul**, M. Baryakhtar, J. Huang, and K.K. Berggren, "Dark Matter Haloscope with a Disordered Dielectric Absorber", [Physical Review D](#) (2026).
4. **D.J. Paul**, T.X. Zhou, and K.K. Berggren, "Photolithography Compatible Three-Terminal Superconducting Switch for Driving CMOS Loads", [Physical Review Applied](#) (2025).
5. **D.J. Paul**, T.X. Zhou and K.K. Berggren, "Determination of Mid-Infrared Refractive Indices of Superconducting Thin Films Using Fourier Transform Infrared Spectroscopy", [Applied Physics Letters](#) (2025).
6. M.I. Tahmid, **D.J. Paul**, and M.Z. Baten, "Emergence and Tunability of Transmission Gap in the Strongly Disordered Regime of a Dielectric Random Scattering Medium", [Optics Express](#), OSA Publishing (2021).

7. **D.J. Paul**, S. Hossain, and M.Z. Baten, "Disorder Induced Rotational-Symmetry Breaking to Control Directionality of Whispering Gallery Modes in Circularly Symmetric Nanowire Assembly", *Optics Express*, OSA Publishing (2019).
8. **D.J. Paul**, A.A. Mimi, A. Hazari, P. Bhattacharya, and M.Z. Baten, "Finite-Difference Time-Domain Analysis of the Tunability of Anderson Localization of Light in Self-Organized GaN Nanowire Arrays", *Journal of Applied Physics*, AIP Publishing (2019).

Under Preparation:

1. **D.J. Paul**, D.F. Santavicca, K.K. Berggren, *et al.*, "Dark Count Events in Thin-Film YBCO Superconducting Nanowires at Temperatures up to 50 K".

INVITED AND CONFERENCE TALKS

- Applied Superconductivity Conference, Pittsburgh, Pennsylvania 09/2026
"Kinetic inductance and relaxation oscillation dynamics in thin-film $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ nanowires"
- BREAD Collaboration Workshop, Harvard University 01/2026
"Mid-infrared SNSPDs for Exploring Dark Matter" Host: Prof. Stefan Knirck
- RLE Shannon Fellowship TED Talk, Weston Middle School 12/2025
"Beyond What Our Eyes Can See" Host: Jaye Goldstein
- Applied Superconductivity Conference, Honolulu, Hawaii 10/2022
"Optimizing Infrared Sensitivity of Superconducting Nanowire Single-Photon Detectors"

RESEARCH GRANT WRITING EXPERIENCE

1. DOE QIS-Enabled High Energy Physics (QuantISED 2.0) with Prof. L. Winslow, Prof. K. Berggren, Prof. M. Baryakhtar, Prof. G. Rybka, and Prof. R. Maruyama; contributed to proposal development and wrote one thrust on mid-infrared SNSPD development for the proposed pathfinder experiments. [awarded in 12/2024; [news coverage](#)]
2. NASA APRA ROSES 2023 with B. Korzh and Prof. K. Berggren; wrote one thrust on antenna coupling to improve mid-infrared absorption efficiency in SNSPDs. [awarded in 06/2024]

RESEARCH EXPERIENCE

- Research Assistant, Quantum Nanostructures and Nanofabrication Group, MIT
Advisor: Prof. Karl Berggren 01/2021 – present
Superconducting device measurements and superconducting nanowire single-photon detector (SNSPD) characterization at cryogenic temperatures (300 mK to 4 K); conducted fundamental research to improve the mid-IR sensitivity and increase the operating temperature of SNSPDs.
- Research Assistant, Solid-state Electronics and Photonics Group, BUET
Advisor: Prof. Md Zunaid Baten 03/2017 – 06/2020
Developed FDTD simulation framework to study light-matter interactions in nanoscale devices.
- Research Assistant, Nano Device Research Group, BUET
Advisor: Prof. Quazi D. M. Khosru 03/2016 – 02/2017
Developed a self-consistent charge-transport simulation model for tunneling FET.

TEACHING AND MENTORSHIP EXPERIENCE

- Teaching Assistant, 6.2540: [Nanotechnology: From Atoms to Systems](#), MIT EECS — Fall 2023
Applied quantum mechanics and nanofabrication techniques (undergraduate).
- Lecturer, Bangladesh University of Engineering and Technology 05/2017 – 08/2020
Courses taught: Electronic Circuits, Digital Electronics, and VLSI.
- Undergraduate Research Mentor (UROP), MIT EECS 2024–Present
Mentored three MIT undergraduates on superconducting device simulations and cryogenic measurements.